Language Models and Agents: Individual Project Report - Bonus

Korlapati Keerthana
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1 Bonus: Positional-Embedding Ablation

The optional positional-embedding ablation was performed for (Model H) Telugu. The ablated model removes positional embeddings while retaining the rest of the Transformer architecture.

Table 1: Telugu positional-embedding ablation.

Metric	**Standard**	**No position**
Parameters	23.04M	22.81M
Test loss	3.124	3.171
Perplexity	22.73	23.83
BPB	0.467	0.474

Removing positional embeddings increases test perplexity from 22.73 to 23.83 and BPB from 0.467 to 0.474. This is consistent with the role of positional information in a decoder-only Transformer: without explicit position information, self-attention does not directly represent token order.

1.1 Bonus Generation Evaluation

The full generation evaluation was also run for the no-position model.

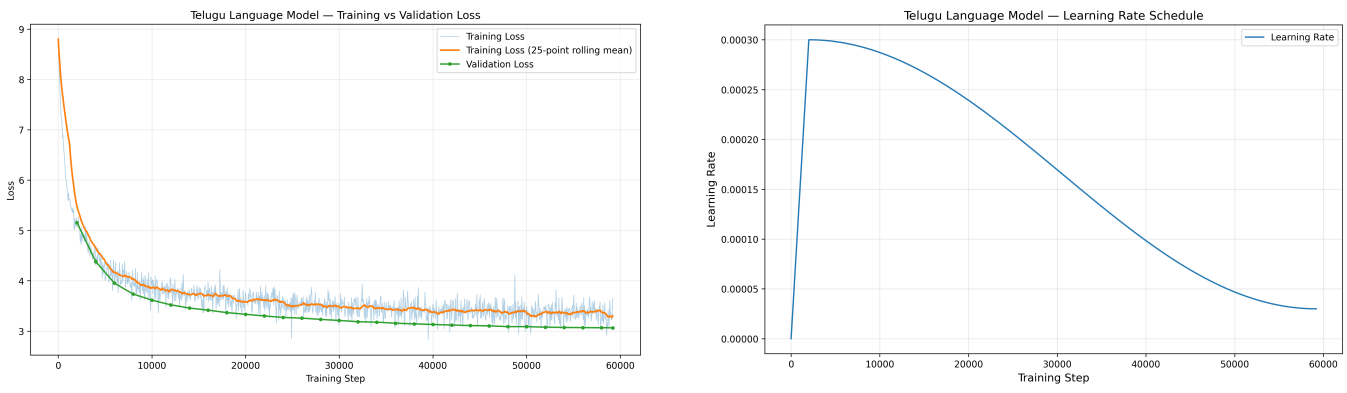
Table 2: Telugu no-position generation metrics.

Setting	BLEU	chrF	ROUGE-L	Rep.	D-1	D-2
Greedy	8.66	23.16	.141	.558	.288	.392
$T = 0.5$	7.50	23.52	.133	.264	.416	.634
$T = 1.0$	4.78	22.50	.089	.016	.732	.961
$T = 1.5$	.55	17.22	.025	.000	.811	.998

The generation results are mixed compared with the standard model. Some reference-overlap metrics are slightly higher for the ablation, but this does not change the intrinsic language-modeling result. Repetition is slightly higher for greedy decoding and Distinct-1 is lower.

The same temperature trend is still visible: higher temperature reduces repetition and increases diversity, while very high temperature reduces reference overlap.

1.2 Bonus Training Curve



(a) Training/validation loss

(b) Learning-rate schedule

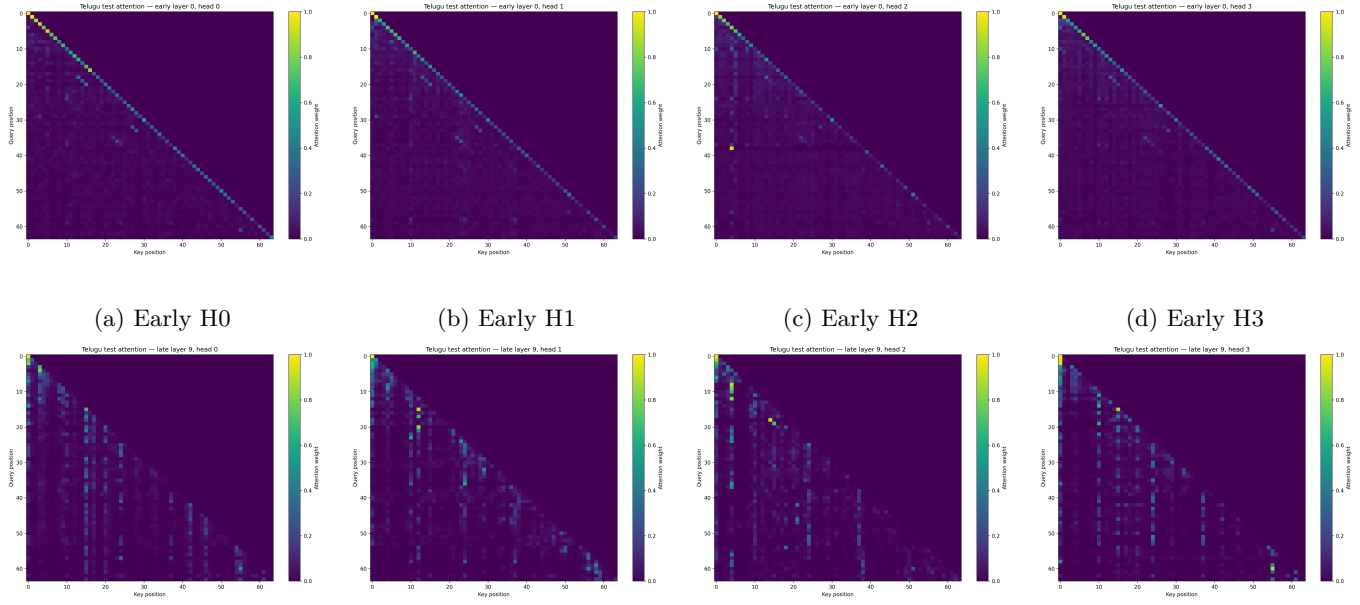
Figure 1: Training results for the Telugu no-position ablation.

1.3 Bonus Attention Analysis

The no-position model was also evaluated with the same attention analysis used for the standard model.

Table 3: Telugu no-position attention statistics.

Metric	**Standard**	**No position**
Mean attention entropy	2.187	2.304
Mean attention distance	11.78	11.45



(a) Early H0 (b) Early H1 (c) Early H2 (d) Early H3

(e) Late H0 (f) Late H1 (g) Late H2 (h) Late H3

Figure 2: Attention maps for the Telugu no-position ablation.

The no-position model has slightly higher average attention entropy and slightly lower mean attention distance than the standard model. The heatmaps still show different attention patterns across heads, although the model no longer receives an explicit positional embedding.